



Yakeen Plus (2025)

Practice Test - 05

DURATION : 200 Minutes

DATE : 16/10/2024

M. MARKS : 720

Topics Covered

Physics :	Thermal Properties of matter, Kinetic Theory, Thermodynamics
Chemistry :	Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Biology :	(Botany): Plant Kingdom, Morphology of Flowering Plants (Zoology): Locomotion & Movement

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

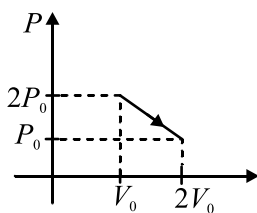
OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

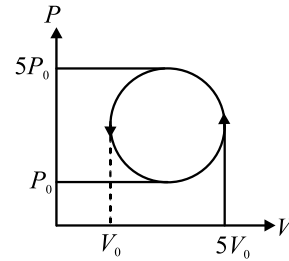
SECTION - A

- A black body, which is at a high temperature T K, thermal radiation emitted at the rate of E W/m². If the temperature falls to $T/4$ K, the thermal radiation emitted in W/m² will be:
 - E
 - $\frac{E}{4}$
 - $\frac{E}{64}$
 - $\frac{E}{256}$
- The coefficient of volume expansion of glycerine is $49 \times 10^{-5} \text{ K}^{-1}$. The fractional change in the density on a 30°C rise in temperature is:
 - 1.47×10^{-2}
 - 1.47×10^{-3}
 - 1.47×10^{-1}
 - 1.47×10^{-4}
- A sample of an ideal gas undergoes an isothermal expansion. If dQ , dU and dW represent the amount of heat supplied, the change in internal energy and the work done respectively. then
 - $dQ = +ve, dU = +ve, dW = +ve$
 - $dQ = +ve, dU = 0, dW = +ve$
 - $dQ = +ve, dU = +ve, dW = 0$
 - $dQ = -ve, dU = -ve, dW = -ve$
- Select correct statement related to heat.
 - Heat is possessed by a body
 - Hot water contains more heat as compared to cold water
 - Heat is a energy which flows due to temperature differences
 - All of these
- A monatomic gas contained in vessel is subjected to a thermodynamic process such that its pressure changes with volume as shown in graph, then change in internal energy during process will be:



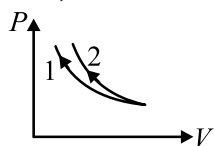
- Zero
- $3P_0V_0$
- $3/2P_0V_0$
- $5P_0V_0$

- The work done during the cyclic process shown in the diagram is:

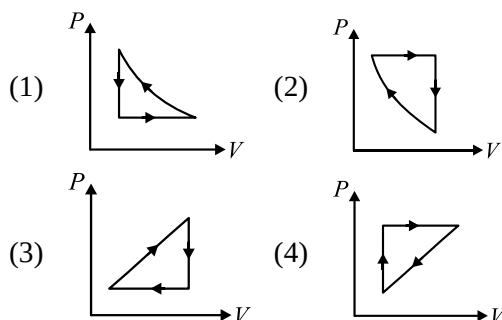
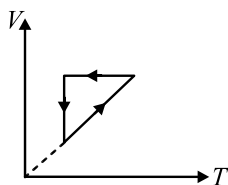


- $4\pi P_0V_0$
 - $-4\pi P_0V_0$
 - $-\pi P_0V_0$
 - πP_0V_0
- A monatomic gas at a pressure P , having a volume V expands isothermally to a volume $3V$ and then compresses adiabatically to the volume V , the final pressure of the gas is:
 - $3^{\frac{5}{3}}P$
 - $3^{\frac{2}{3}}P$
 - $\frac{P}{3}$
 - $3P$
 - In the adiabatic expansion of an ideal gas, select the **incorrect** statement.
 - There is decrease in temperature of the gas
 - There is decrease in internal energy of the gas
 - The work done is positive
 - The work done by gas is equal to heat supplied
 - An ideal gas of adiabatic exponent $\left(\gamma = \frac{7}{5}\right)$ is expanding at constant pressure. The ratio of $dQ : dU : dW$ is (Symbols have their usual meanings):
 - $7 : 5 : 1$
 - $7 : 2 : 5$
 - $2 : 1 : 1$
 - $7 : 5 : 2$
 - Which of the following is **incorrect**?
 - If two systems A and B are in thermal equilibrium with system C separately, then A and B will be in thermal equilibrium
 - Zeroth law of thermodynamics defines temperature
 - Temperature does not determine the direction of flow of heat when two bodies are placed in thermal contact
 - Internal energy of an ideal gas depends on the state of the system

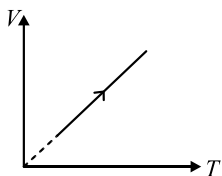
11. For the two adiabatics shown on a pressure-volume ($P - V$) diagram, if process 1 is for oxygen gas, then process 2 may be for (Assume all gases are ideal):



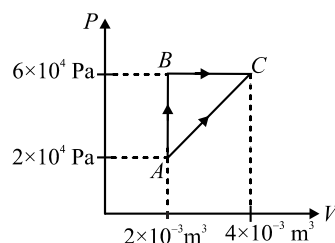
- (1) N_2
 (2) CO_2
 (3) He
 (4) O_3
12. If a thermodynamic cyclic process is as shown on volume-temperature ($V - T$) diagram, then on pressure - volume ($P - V$) diagram it will be shown as:



13. A monoatomic ideal gas undergoes a thermodynamic process as shown in the figure. The molar specific heat of the process is:



- (1) $\frac{3}{2}R$ (2) $\frac{5}{2}R$
 (3) $\frac{7}{5}R$ (4) $2R$
14. Figure below shows two paths that may be taken by a gas to go from a state A to a state C. In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be:



- (1) 300 J (2) 380 J
 (3) 500 J (4) 460 J
15. If a gas is compressed adiabatically:
 (1) the internal energy of the gas increases.
 (2) internal energy of the gas decreases.
 (3) internal energy of the gas does not change.
 (4) work done by the gas is positive.
16. PV plots of two gases during adiabatic processes are shown. Plot 1 and 2 should correspond respectively to:
-
- (1) He and Ne (2) N_2 and Ne
 (3) Ne and N_2 (4) O_2 and N_2
17. By what percentage should the pressure of a given mass of a gas be increased so as to decrease its volume by 10% at a constant temperature?
 (1) 10% (2) 9.1%
 (3) 11.1% (4) 12%
18. The volume of a certain mass of gas at constant pressure is doubled to its value at $0^\circ C$. The temperature of the gas will be:
 (1) $173^\circ C$ (2) 273 K
 (3) $273^\circ C$ (4) 173 K
19. The rms speed of a molecule of oxygen at $127^\circ C$ is half that of a molecule of hydrogen at:
 (1) 100 K (2) 273 K
 (3) 173 K (4) $100^\circ C$
20. Five molecules of a gas have speeds 1, 2, 3, 4 and 5 km/s. The ratio of rms speed to the average speed of the gas molecule will be:
 (1) 10 : 3 (2) $\sqrt{11} : 3$
 (3) 3 : 10 (4) $\sqrt{3} : \sqrt{10}$
21. A container has 64 g of O_2 at temperature T and pressure P . If same container has 8 g of H_2 at temperature $2T$, then its pressure will be:
 (1) P (2) $2P$
 (3) $4P$ (4) $8P$

22. Two perfect gases at absolute temperature T_1 and T_2 are mixed in a closed vessel. Their number of moles is N_1 and N_2 respectively. Assuming no heat loss to the surroundings, the final temperature of the mixture will be:

- (1) $\frac{T_1 + T_2}{2}$
- (2) $\frac{N_1 T_1 + N_2 T_2}{2}$
- (3) $\sqrt{T_1 T_2}$
- (4) $\frac{N_1 T_1 + N_2 T_2}{N_1 + N_2}$

23. In a vessel, the gas is at a pressure P . If the mass of all the molecules is halved and their speed is doubled, then the resultant pressure will be:

- (1) $4P$
- (2) $2P$
- (3) P
- (4) $P/2$

24. Three perfect gases at absolute temperatures T_1 , T_2 and T_3 are mixed. If number of molecules of the gases are n_1 , n_2 and n_3 respectively then temperature of mixture will be (assume no loss of energy):

- (1) $\frac{T_1 + T_2 + T_3}{3}$
- (2) $\frac{n_1^2 T_1 + n_2^2 T_2 + n_3^2 T_3}{n_1 + n_2 + n_3}$
- (3) $\frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$
- (4) $\frac{T_1 + T_2 + T_3}{n_1 + n_2 + n_3}$

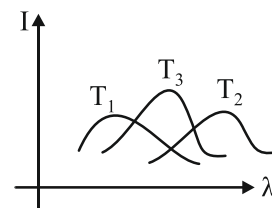
25. The ratio of average translatory kinetic energy of He gas molecules to O_2 gas molecules:

- (1) $\frac{25}{21}$
- (2) $\frac{21}{25}$
- (3) $\frac{3}{2}$
- (4) 1

26. In a cyclic process, work done by the system is:

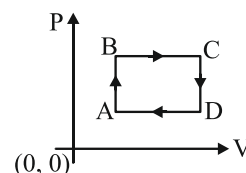
- (1) zero.
- (2) equal to heat given to the system.
- (3) more than the heat given to system.
- (4) independent of heat given to the system.

27. The plots of intensity versus wavelength for three black bodies at temperatures T_1 , T_2 and T_3 respectively are as shown. Their temperature are such that:



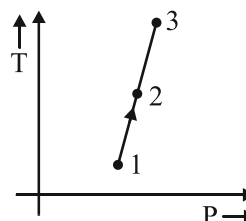
- (1) $T_1 > T_2 > T_3$
- (2) $T_1 > T_3 > T_2$
- (3) $T_2 > T_3 > T_1$
- (4) $T_3 > T_2 > T_1$

28. The figure shows P - V diagram of a thermodynamic cycle. Which corresponding curve is **correct**?



- (1)
- (2)
- (3)
- (4)

29. A sample of gas goes through a process shown in graph. Arrange the volume of given points in increasing order:

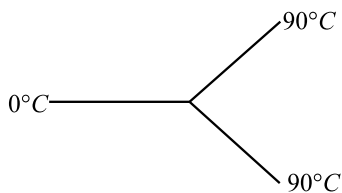


- (1) $V_1 > V_2 > V_3$
- (2) $V_1 > V_2 = V_3$
- (3) $V_3 > V_2 > V_1$
- (4) $V_3 > V_2 = V_1$

30. The temperature of 100 gm of water is to be raised from 24°C to 90°C by adding steam to it. The mass of the steam required for this purpose is:

- (1) 20 g
- (2) 15 g
- (3) 12 g
- (4) 18 g

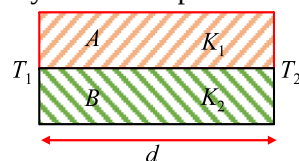
31. Three rods made of the same material and having the same cross-section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of the junction of the three rods will be:



- (1) 45°C (2) 60°C
 (3) 30°C (4) 20°C
32. Choose the **correct** equation for interconversion of temperature scales.
- (1) $\frac{T_C - 0}{100} = \frac{T_F - 32}{180}$
 (2) $\frac{T_F - 32}{180} = \frac{T_K + 273.15}{100}$
 (3) $\frac{T_F - 32}{180} = \frac{T_K + 273.15}{180}$
 (4) $\frac{T_C - 0}{180} = \frac{T_F - 32}{100}$
33. Heat content of a body depends on:
- (1) mass of the body.
 (2) temperature of the body.
 (3) specific heat capacity.
 (4) All of the above
34. A block of ice floats on a liquid of density 1.2 in a beaker then level of liquid when ice completely melt:
- (1) Remains same (2) Rises
 (3) Lowers (4) Either (2) or (3)
35. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: It is hotter over the top of a fire than at the same distance on the sides.
Reason R: Air surrounding the fire conducts more heat upwards.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

SECTION - B

36. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Blue star is at high temperature than red star.
Reason R: Wein's displacement law states that $T \propto (1/\lambda_m)$.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
37. Two sphere made of same substance have diameters in the ratio 1 : 2. Their thermal capacities are in the ratio of:
- (1) 1 : 2 (2) 1 : 8
 (3) 1 : 4 (4) 2 : 1
38. Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2 . The thermal conductivity of the composite rod will be:



- (1) $\frac{3(K_1 + K_2)}{2}$
 (2) $K_1 + K_2$
 (3) $2(K_1 + K_2)$
 (4) $\frac{K_1 + K_2}{2}$
39. The thermal capacities of two bodies are in the ratio of 1 : 8. If the rate of loss of heat are equal for the two bodies under identical conditions of surroundings, then the ratio of rate of fall of temperature of the two bodies is:
- (1) 1 : 4
 (2) 4 : 1
 (3) 1 : 8
 (4) 8 : 1
40. The original temperature of a black body is 727°C . Calculate temperature at which total radiant energy from this black body becomes double: ($2^{1/4} = 1.19$):
- (1) 971 K (2) 1190 K
 (3) 2001 K (4) 1458 K

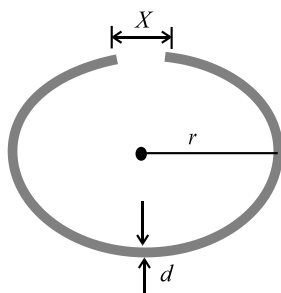
41. The rate of dissipation of heat by a black body at temperature T is Q . What will be the rate of dissipation of heat by another body at temperature $2T$ and emissivity 0.25 ?

(1) $16Q$
 (2) $4Q$
 (3) $8Q$
 (4) $4.5Q$

42. When water is heated from 0°C to 10°C , then its density?

(1) Increases
 (2) Decreases
 (3) First increases then decreases
 (4) First decreases then increases

43. A cylindrical metal rod of length L_0 is shaped into a ring with a small gap as shown. On heating the system:



(1) x decreases, r and d increase
 (2) x and r increase, d decreases
 (3) x , r and d all increase
 (4) Data insufficient to arrive at a conclusion

44. In making an alloy, a substance of specific gravity s_1 and mass m_1 is mixed with another substance of specific gravity s_2 and mass m_2 ; then the specific gravity of the alloy is:

(1) $\left(\frac{m_1 + m_2}{s_1 + s_2} \right)$
 (2) $\left(\frac{s_1 s_2}{m_1 + m_2} \right)$
 (3) $\frac{m_1 + m_2}{\left(\frac{m_1}{s_1} + \frac{m_2}{s_2} \right)}$
 (4) $\frac{\left(\frac{m_1}{s_1} + \frac{m_2}{s_2} \right)}{m_1 + m_2}$

45. On a new scale of temperature (which is linear) and called the W scale, the freezing and boiling points of water are $39^\circ W$ and $239^\circ W$

respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale?

(1) $139^\circ W$
 (2) $78^\circ W$
 (3) $117^\circ W$
 (4) $200^\circ W$

46. There is a hole in the bottom of tank having water. If total pressure at bottom is 3 atm ($1 \text{ atm} = 10^5 \text{ N/m}^2$) then the velocity of water flowing from hole is:

(1) $\sqrt{400} \text{ m/s}$
 (2) $\sqrt{600} \text{ m/s}$
 (3) $\sqrt{60} \text{ m/s}$
 (4) None of these

47. The temperature at which Fahrenheit reading is five times of Celsius reading is:

(1) 10°C
 (2) 12°C
 (3) 18°C
 (4) -10°C

48. Coefficient of linear expansion of brass and steel rods are α_1 and α_2 . Lengths of brass and steel rods are ℓ_1 and ℓ_2 respectively. If $(\ell_2 - \ell_1)$ is maintained same at all temperatures, which one of the following relations holds good?

(1) $\alpha_1 \ell_1 = \alpha_2 \ell_2$
 (2) $\alpha_1 \ell_2 = \alpha_2 \ell_1$
 (3) $\alpha_1 \ell_2^2 = \alpha_2 \ell_1^2$
 (4) $\alpha_1^2 \ell_2 = \alpha_2^2 \ell_1$

49. If temperature of an object is 140°F , then its temperature in centigrade is:

(1) 105°C
 (2) 32°C
 (3) 140°C
 (4) 60°C

50. 5 g of ice at 0°C is dropped in a beaker containing 20 g of water at 40°C . The final temperature will be:

(1) 32°C
 (2) 16°C
 (3) 8°C
 (4) 24°C

SECTION-(II) CHEMISTRY**SECTION - A**

51. The compound containing coordinate bond is:

- (1) O_3 (2) BF_3
(3) PCl_5 (4) All of these

52. Match **List-I** with **List-II**.

List-I (Compound)		List-II (Geometry/Shape)	
(A)	XeF_6	(I)	Distorted octahedral
(B)	XeO_3	(II)	Square planar
(C)	$XeOF_4$	(III)	Pyramidal
(D)	XeF_4	(IV)	Square pyramidal

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
(2) A-I, B-II, C-IV, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-IV, B-I, C-II, D-III

53. Given below are two statements:

Statement I: Electronic Geometry of NH_3 molecule is tetrahedral.

Statement II: Nitrogen is sp^3 hybridized in NH_3 .

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

54. Arrange the following ions in the order of decreasing X–O bond length, where X is the central atom:

- (1) $ClO_4^- > SO_4^{2-} > PO_4^{3-} > SiO_4^{4-}$
(2) $SiO_4^{4-} > PO_4^{3-} > SO_4^{2-} > ClO_4^-$
(3) $SiO_4^{4-} > PO_4^{3-} > ClO_4^- > SO_4^{2-}$
(4) $SiO_4^{4-} > SO_4^{2-} > PO_4^{3-} > ClO_4^-$

55. A π -bond is formed by the overlap of:

- (1) s-s orbitals.
(2) s-p orbitals.
(3) p-p orbitals in head on head manner.
(4) p-p orbitals in sideways manner.

56. Given below are two statements:

Statement I: SO_2 and H_2O both are V-shaped molecules.

Statement II: The bond angle of SO_2 is less than that of H_2O .

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

57. What is the formal charge on hydrogen atom in water molecule?

- (1) 0 (2) $-\frac{1}{2}$
(3) 1 (4) $+\frac{1}{2}$

58. Which of the following molecules has zero dipole moment?

- (1) CO_2
(2) BF_3
(3) SO_3
(4) All of these

59. According to Fajan's rule, covalent character is favoured by:

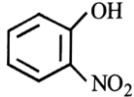
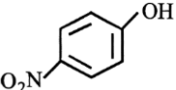
- (1) small cation, large anion
(2) small anion, small cation
(3) large anion, large cation
(4) small anion, large cation

60. Among the following molecules

- (i) XeO_3
(ii) $XeOF_4$
(iii) XeF_6

Pair of molecules having same number of lone pairs on Xe are:

- (1) (i) and (ii) only
(2) (i) and (iii) only
(3) (ii) and (iii) only
(4) (i), (ii) and (iii)

61. The types of hybridisation of the five carbon atoms from left to right in the molecule $\text{CH}_3 - \text{CH} = \text{C} = \text{CH} - \text{CH}_3$ are:
 (1) $\text{sp}^3, \text{sp}^2, \text{sp}^2, \text{sp}^2, \text{sp}^3$
 (2) $\text{sp}^3, \text{sp}, \text{sp}^2, \text{sp}^2, \text{sp}^3$
 (3) $\text{sp}^3, \text{sp}^2, \text{sp}, \text{sp}^2, \text{sp}^3$
 (4) $\text{sp}^3, \text{sp}^2, \text{sp}^2, \text{sp}, \text{sp}^3$
62. Which of the following species show deviation from the octet's rule?
 (1) AlCl_3 (2) SF_6
 (3) NO (4) All of these
63. The species containing three bond pairs and zero lone pair of electrons around the central atom is:
 (1) NH_2^- (2) ClF_3
 (3) H_2S (4) BF_3
64. Among LiCl , RbCl , BeCl_2 and MgCl_2 , the compound with greatest and least ionic character respectively are:
 (1) LiCl , RbCl (2) RbCl , MgCl_2
 (3) RbCl , BeCl_2 (4) MgCl_2 , BeCl_2
65. Which of the following compound shows highest lattice energy?
 (1) LiF (2) LiCl
 (3) LiBr (4) LiI
66. The strength of bonds by $2s-2s$, $2p-2p$ and $2p-2s$ overlapping has the order:
 (1) $2s-2s > 2p-2p > 2s-2p$
 (2) $2s-2s > 2p-2s > 2p-2p$
 (3) $2p-2p > 2s-2p > 2s-2s$
 (4) $2p-2p > 2s-2s > 2p-2s$
67. According to molecular orbital theory for O_2^+ :
 (1) O_2^+ is paramagnetic and bond order is less than O_2 .
 (2) O_2^+ is paramagnetic and bond order is more than O_2 .
 (3) O_2^+ is diamagnetic and bond order is less than O_2 .
 (4) O_2^+ is diamagnetic and bond order is more than O_2 .
68. Which one of the following species has trigonal planar shape?
 (1) N_3^- (2) NO_3^-
 (3) ClF_3 (4) CO_2
69. The number of $d\pi - p\pi$ bonds present respectively in SO_2 , SO_3 and ClO_4^- are:
 (1) 0, 1, 2
 (2) 1, 2, 3
 (3) 2, 3, 4
 (4) 2, 3, 3
70. The change in hybridisation (if any) of the 'Al' atom in the following reaction is:
 $\text{AlCl}_3 + \text{Cl}^- \rightarrow \text{AlCl}_4^-$
 (1) sp^3 to sp^2
 (2) sp^2 to sp^3
 (3) sp^3 to sp^3d
 (4) no change
71. The compound which contains both ionic and covalent bonds is:
 (1) CH_4 (2) H_2
 (3) KCN (4) KCl
72. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: ClF_3 is T-shape molecule.
Reason R: It has two lone pairs arranged at 180° angle to each other.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
73. The vapour pressure of  is higher than  due to:
 (1) dipole moment.
 (2) dipole-dipole interaction.
 (3) Hydrogen bonding.
 (4) lattice structure.
74. If 'z' is the internuclear axis, orbital ' p_z ' and ' p_z ' overlap with each other to form:
 (1) π bond
 (2) σ bond
 (3) δ bond
 (4) No bond will be formed

75. Which of the following represents **correctly** the order of filling of the molecular orbitals of B_2 , C_2 , N_2

- (1) $(\pi_{2p_x} = \pi_{2p_y}) < \sigma_{2p_z} < (\pi_{2p_y}^* = \pi_{2p_x}^*) < \sigma_{2p_z}^*$
- (2) $\sigma_{2p_z} < (\pi_{2p_x} = \pi_{2p_y}) < (\pi_{2p_x}^* = \pi_{2p_y}^*) < \sigma_{2p_z}^*$
- (3) $(\pi_{2p_z} = \pi_{2p_x} = \sigma_{2p_y}) < (\pi_{2p_x}^* = \pi_{2p_y}^*) < \sigma_{2p_z}^*$
- (4) $\pi_{2p_x} < \pi_{2p_y} < \sigma_{2p_z} > \pi_{2p_y}^* < (\pi_{2p_z}^* = \sigma_{2p_z}^*)$

76. Consider the following changes



The energy required to pull out the two electrons are E_1 and E_2 respectively. The **correct** relationship between two energies would be:

- (1) $E_1 < E_2$
- (2) $E_1 = E_2$
- (3) $E_1 > E_2$
- (4) $E_1 \geq E_2$

77. Which one of the following is the reason behind fluorine has lower electron affinity than chlorine?

- (1) Smaller radius of fluorine, high electron density
- (2) Smaller radius of chlorine, high electron density
- (3) Bigger radius of fluorine, less electron density
- (4) Smaller radius of chlorine, less electron density

78. The first ionization potential (in eV) of Be and B, respectively are:

- (1) 8.29, 9.32
- (2) 9.32, 9.32
- (3) 8.29, 8.29
- (4) 9.32, 8.29

79. The element Uuq will belong to which of the following family and what will be its electronic configuration?

- (1) Carbon family, $[Rn]5f^{14} 6d^{10} 7s^2 7p^2$
- (2) Oxygen family, $[Rn]5f^{14} 6d^{10} 7s^2 7p^4$
- (3) Nitrogen family, $[Rn]5f^{14} 6d^{10} 7s^2 7p^3$
- (4) Halogen family, $[Rn]5f^{14} 6d^{10} 7s^2 7p^5$

80. Consider the following elements:

Cl, F, S, P

The **correct** statement regarding the above elements is:

- (1) Sulphur has the most negative electron gain enthalpy among the given elements.
- (2) Adding an electron to 3p-orbital leads to greater repulsion than adding an electron to 2p-orbital.
- (3) Phosphorus has the least negative electron gain enthalpy among the given elements.
- (4) As we move from left to right across a period, the electron gain enthalpy generally becomes less negative.

81. The atomic radius of 'P' is 0.11 nm. The atomic radius of Cl will be:

- (1) Smaller than P
- (2) Greater than P
- (3) Same as P
- (4) Twice of P

82. The **incorrect** statement among the following is:

- (1) The first ionisation enthalpy of Al is less than the first ionisation enthalpy of Mg.
- (2) The second ionisation enthalpy of Mg is greater than the second ionisation enthalpy of Na.
- (3) The first ionisation enthalpy of Na is less than the first ionisation enthalpy of Mg.
- (4) The third ionisation enthalpy of Mg is greater than the third ionisation enthalpy of Al.

83. Given below are two statements:

Statement I: Noble gases have positive electron gain enthalpy.

Statement II: Halogens have negative electron gain enthalpy.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

84. Among the following the most electronegative element is:

- (1) Li
- (2) Mg
- (3) H
- (4) Na

85. Screening effect is **not** observed in:

- (1) He^+
- (2) Li^{2+}
- (3) Be^{3+}
- (4) In all the three

SECTION - B

86. Which of the following is the **incorrect** match for the position of the element in the periodic table?

- (1) $[Ar]3d^5 4s^1 \Rightarrow 4^{th}$ period, 6^{th} group
- (2) $[Kr]4d^{10} 5s^2 \Rightarrow 5^{th}$ period, 10^{th} group
- (3) $[Rn]5f^0 6d^2 7s^2 \Rightarrow 7^{th}$ period, 3^{th} group
- (4) $[Xe]4f^{14} 5d^2 6s^2 \Rightarrow 6^{th}$ period, 4^{th} group

87. The shapes of IF_5 and IF_7 are respectively:
- (1) square pyramidal and pentagonal bipyramidal.
 - (2) octahedral and pyramidal.
 - (3) trigonal bipyramidal and square pyramidal.
 - (4) distorted square planar and distorted octahedral.

88. The bond order and the magnetic characteristics of N_2 are:

- (1) 3, paramagnetic
- (2) $2\frac{1}{2}$, diamagnetic
- (3) 3, diamagnetic
- (4) $2\frac{1}{2}$, paramagnetic

89. The **correct** order of radii is:

- (1) $\text{Li} < \text{Na} > \text{K}$
- (2) $\text{Li} < \text{Be} < \text{B}$
- (3) $\text{Fe}^{3+} < \text{Fe}^{4+} < \text{Fe}^{2+}$
- (4) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$

90. Which of the following sequence **correctly** represents the decreasing acidic nature of oxides?

- (1) $\text{Li}_2\text{O} > \text{BeO} > \text{B}_2\text{O}_3 > \text{CO}_2 > \text{N}_2\text{O}_3$
- (2) $\text{N}_2\text{O}_3 > \text{CO}_2 > \text{B}_2\text{O}_3 > \text{BeO} > \text{Li}_2\text{O}$
- (3) $\text{CO}_2 > \text{N}_2\text{O}_3 > \text{B}_2\text{O}_3 > \text{BeO} > \text{Li}_2\text{O}$
- (4) $\text{B}_2\text{O}_3 > \text{CO}_2 > \text{N}_2\text{O}_3 > \text{Li}_2\text{O} > \text{BeO}$

91. Which of the following is **true** about the element $_{33}\text{As}$ according to Modern periodic table:

- (1) it is a 5th period element.
- (2) it is a p-block element.
- (3) it belongs to 16th group.
- (4) it is transition metal.

92. Match **List-I** with **List-II**.

List-I		List-II	
(A)	He	(I)	Highest negative electron gain enthalpy
(B)	Cl	(II)	Most electropositive element
(C)	Cs	(III)	Stronger reducing agent
(D)	Li	(IV)	Highest ionisation enthalpy

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-I, C-II, D-III

93. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Atomic radius of gallium is higher than that of aluminium.

Reason R: The presence of additional 10 d-electrons offer poor screening effect for the outer electrons from increased nuclear charge in gallium.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

94. The size of isoelectronic species; F^- , Na^+ is affected by:

- (1) effective nuclear charge (Z_{eff}).
- (2) valence principal quantum number (n).
- (3) electron-electron interaction in the outer orbitals.
- (4) none of the factors because their size is the same.

95. The **correct** increasing order of atomic radii among the C, N, O, F is:

- (1) $\text{F} < \text{C} < \text{N} < \text{O}$
- (2) $\text{F} < \text{N} < \text{O} < \text{C}$
- (3) $\text{O} < \text{F} < \text{N} < \text{C}$
- (4) $\text{F} < \text{O} < \text{N} < \text{C}$

96. The highest occupied MO in N_2 and O_2^+ respectively are (take x-axis as internuclear axis)

- (1) $\sigma 2p_x$, $\pi^* 2p_y$
- (2) $\pi 2p_y$, $\pi 2p_z$
- (3) $\sigma^* 2p_x$, $\sigma 2p_x$
- (4) $\pi^* 2p_y$, $\pi^* 2p_z$

97. The f-block elements belong to which group of the periodic table?

- (1) 2
- (2) 3
- (3) 4
- (4) 5

98. The element expected to form largest ion to achieve the nearest noble gas configuration is:

- (1) Na
- (2) O
- (3) F
- (4) N

99. Higher is the bond order, greater is:

- (1) Bond dissociation energy
- (2) Covalent character
- (3) Bond length
- (4) Paramagnetism

100. H—O—H bond angle in H_2O is:

- (1) less than $109^\circ 28'$
- (2) more than $109^\circ 28'$
- (3) equal to $109^\circ 28'$
- (4) less than or equal to $109^\circ 28'$

SECTION-(III) BOTANY

SECTION - A

101. Read the given characters and identify the plant family.

- A. Pentamerous flowers with epicalyx.
- B. Corolla with mucilage.
- C. Androecium with numerous stamens.
- D. Pentacarpellary, syncarpous, superior ovary.
- E. axile placentation.

- (1) Fabaceae.
- (2) Asteraceae.
- (3) Poaceae.
- (4) Malvaceae.

102. Placentation in Fabaceae family is:

- (1) marginal. (2) axile.
- (3) basal. (4) parietal.

103. Given below are two statements:

Statement I: Bryophytes are amphibians of the plant kingdom.

Statement II: Bryophytes can live in soil but are dependent on water for sexual reproduction.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

104. In pheophyceae, the plant body is usually attached to the substratum by a (A), and has a stalk, the (B) and leaf like photosynthetic organ-the (C).

- | (A) | (B) | (C) |
|--------------|----------|----------|
| (1) holdfast | stipe | frond |
| (2) stipe | frond | holdfast |
| (3) frond | stipe | holdfast |
| (4) stipe | holdfast | frond |

105. Select the **correct** statements w.r.t. Asteraceae family.

- A. Flowers are reduced to florets.
- B. Calyx is modified to hairs or pappus.
- C. Flowers are actinomorphic (ray florets) or zygomorphic (disc florets).

D. Ray florets are found towards the periphery (they are younger) and disc florets are distributed near the center (thus older).

choose the most appropriate answer from the options given below:

- (1) A and B only. (2) A, B and C only.
- (3) A, B and D only. (4) C and D only.

106. Floridean starch is very similar to A in structure and found in B.

- (1) A - amylopectin, B - *Gracilaria*
- (2) A - amylopectin, B - *Laminaria*
- (3) A - glycogen, B - *Volvox*
- (4) A - glucose, B - *Sargassum*

107. Root hairs are present in/on:

- (1) region of elongation.
- (2) region of maturation.
- (3) region of meristematic activity.
- (4) root cap.

108. In pteridophytes, a spore germinates to produce:

- (1) sporophyte.
- (2) sporangia.
- (3) prothallus.
- (4) microsporophyll.

109. Match List-I with List-II.

List-I		List-II	
(A)	Brassicaceae	(I)	Marigold
(B)	Poaceae	(II)	China rose
(C)	Asteraceae	(III)	Wheat
(D)	Malvaceae	(IV)	Mustard

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

110. Offsets are found in:

- (1) *Pistia*. (2) mint.
- (3) banana. (4) pineapple.

111. Pyrenoids, the storage bodies located in the chloroplasts of most members of Chlorophyceae contain proteins besides:

- (1) glycogen. (2) starch.
- (3) cellulose. (4) amylopectin.

- 112.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Inflorescence in the members of family Brassicaceae is said to be racemose.
Reason R: In the members of family Brassicaceae, floral axis continues to grow and the flowers are borne laterally in an acropetal succession.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true and R is not the correct explanation of A.
- 113.** In liverworts, spores produced on meiosis germinate to form:
- (1) free-living gametophytes.
 - (2) dependent gametophytes.
 - (3) free-living sporophytes.
 - (4) dependent sporophytes.
- 114.** Bryophytes in general are of little economic importance but some **(A)** provide food for herbaceous **(B)**, birds and other animals.
- | (A) | (B) |
|----------------|------------|
| (1) mosses | mammals |
| (2) liverworts | mammals |
| (3) mosses | plants |
| (4) liverworts | plants |
- 115.** Coleorhiza and coleoptile are the protective sheaths covering of ____ and ____ respectively.
- (1) radicle, plumule
 - (2) plumule, radicle
 - (3) plumule, hypocotyl
 - (4) epicotyl, radicle
- 116.** Phycoerythrin, chlorophyll *a* and chlorophyll *d* are characteristics of:
- (1) phaeophyceae.
 - (2) blue green algae.
 - (3) chlorophyceae.
 - (4) rhodophyceae.
- 117.** Choose the **odd** one for the modification of axillary bud.
- (1) Thorn of *Citrus*
 - (2) Tendril of watermelon
 - (3) Spine of *Aloe*
 - (4) Thorn of *Bougainvillea*
- 118.** Palmately compound leaf is the one in which the leaflets develop from a common point i.e. at the tip of:
- (1) rachis.
 - (2) stipule.
 - (3) petiole.
 - (4) nodes.
- 119.** In bryophytes:
- (1) sporophytes are dependent upon gametophytes.
 - (2) sporophytic and gametophytic generations are independent.
 - (3) sporophyte in itself completes the life cycle.
 - (4) gametophytes are dependent upon sporophyte.
- 120.** Which of the following is an **incorrect** match?
- (1) Perigynous flower - Plum, rose and peach
 - (2) Monoadelphous - Pea
 - (3) Hypogynous- Mustard, brinjal
 - (4) Polyadelphous - Citrus
- 121.** Given below are two statements:
Statement I: Each anther is usually bilobed and each lobe has two chambers, the pollen-sacs.
Statement II: A sterile stamen is called staminode.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 122.** Coralloid roots of ____ are association with N_2 – fixing cyanobacteria.
- (1) *Pinus*
 - (2) *Cedrus*
 - (3) *Cycas*
 - (4) *Ginkgo*
- 123.** The ray florets of sunflower has:
- (1) superior ovary.
 - (2) half inferior ovary.
 - (3) half superior ovary.
 - (4) inferior ovary.

124. Which of the following statements is **incorrect**?
- (1) Mosses along with lichens are the first organisms to colonise rocks.
 - (2) *Sphagnum* is used as packing material for transportation of living material.
 - (3) In liverworts, spores are produced after meiosis within the capsule.
 - (4) *Funaria* possesses unicellular unbranched rhizoids.

125. When the margins of sepals or petals overlap one another without any particular direction, the condition is termed as:
- (1) vexillary.
 - (2) imbricate.
 - (3) twisted.
 - (4) valvate.

126. The heterosporous pteridophytes are:
- (1) *Lycopodium* and *Pteris*
 - (2) *Selaginella* and *Psilotum*
 - (3) *Selaginella* and *Salvinia*
 - (4) *Dryopteris* and *Adiantum*

127. Select the **correct** statement w.r.t. mango and coconut.
- (1) They develop from monocarpellary superior ovaries.
 - (2) They develop from monocarpellary inferior ovaries.
 - (3) They have fibrous epicarp.
 - (4) They have fleshy edible mesocarp.

128. Match List-I with List-II.

List-I		List-II	
(A)	Psilopsida	(I)	<i>Psilotum</i>
(B)	Lycopsida	(II)	<i>Equisetum</i>
(C)	Sphenopsida	(III)	<i>Selaginella</i>
(D)	Pteropsida	(IV)	<i>Dryopteris</i>

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
 - (2) A-I, B-III, C-II, D-IV
 - (3) A-IV, B-III, C-II, D-I
 - (4) A-IV, B-I, C-II, D-III
129. Which floral conditions are represented by the symbols $\oplus$ and $\%$ respectively?
- (1) Zygomorphic and actinomorphic flowers
 - (2) Actinomorphic and zygomorphic flowers
 - (3) Hypogynous and epigynous flowers
 - (4) Bisexual and unisexual flowers

130. Which of the following gymnosperms has branched stems?
- (1) *Pinus*
 - (2) *Cycas*
 - (3) *Cedrus*
 - (4) Both (1) and (3)

131. Which one of the following plants shows alternate phyllotaxy?
- (1) Sunflower.
 - (2) Guava.
 - (3) *Alstonia*.
 - (4) *Calotropis*.

132. Identify the **incorrect** match.
- (1) Orchid – Non-endospermic seeds
 - (2) Aleurone layer – Protein rich
 - (3) Scutellum – Cotyledons of dicot seeds
 - (4) Maize – Seed coat fused with fruit wall

133. Select the **incorrect** statement w.r.t. features of Poaceae family.
- (1) Flower is zygomorphic.
 - (2) Each floret has an inferior palea and above it is superior palea.
 - (3) Monocarpellary, unilocular, inferior ovary.
 - (4) Perianth is highly reduced.

134. The sporophyte is differentiated into a foot, seta and capsule in:
- (1) mosses.
 - (2) *Cedrus*.
 - (3) *Cycas*.
 - (4) *Pinus*.

135. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** In pteridophytes, the development of the zygotes into young embryos takes place within the female sporophytes.

Reason R: This event in pteridophytes is a precursor to the seed habit.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is not the correct explanation of A.

SECTION - B

136. Which of the following statements is **incorrect** about leaf?
- (1) A leaf is said to be simple, when its lamina is entire or when incised, the incisions do not touch the midrib.
 - (2) A leaf is said to be compound when the incisions of lamina reach upto the midrib breaking into a number of leaflets.
 - (3) Leaf is the most important vegetative organ for photosynthesis.
 - (4) Leaves originate from shoot apical meristems and are arranged in a basipetal order.

137. The reduced male gametophyte in gymnosperm is called_____.

- (1) microsporangia.
- (2) pollen grain.
- (3) microsporophyll.
- (4) male cone.

138. Radial symmetry is found in flowers of:

- (1) *Cassia*. (2) chilli.
- (3) gulmohur. (4) canna.

139. Given below are two statements:

Statement I: The flower is a modified shoot, meant for sexual reproduction.

Statement II: After fertilisation, the ovary is modified into seed and ovules into flowers.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

140. Leaf base may bear two lateral small leaf like structures called:

- (1) lamina. (2) pulvinus.
- (3) stipules. (4) sepals.

141. Select the **correct** statement w.r.t gymnosperms.

- (1) Gymnosperms are both homosporous and heterosporous.
- (2) *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms.
- (3) *Sequoia* is one of the tallest tree species.
- (4) The leaves of gymnosperms are not well adapted to extremes of climate.

142. Find out the **incorrect** match.

- (1) Stamens united in one bundle - monoadelphous
- (2) Stamens attached to petals – Epipetalous
- (3) Stamens attached to perianth – Diadelphous
- (4) Free stamens – Polyandrous

143. Which kind of placentation is represented by the given figure?



- (1) Marginal (2) Free central
- (3) Parietal (4) Basal

144. Phylogenetic system of classification is based on:

- (1) evolutionary relationships.
- (2) morphological features.
- (3) chemical constituents.
- (4) floral characters.

145. Fruits of which of the given plants have persistent calyx?

- (1) Brinjal
- (2) Mustard
- (3) China rose
- (4) Marigold

146. Find out the **incorrect** match.

- (1) *Ulothrix* — Filamentous algae
- (2) *Chlamydomonas*— Unicellular algae
- (3) *Spirogyra* — Non-flagellate algae
- (4) *Volvox*— Isogamous algae

147. The stony hard part of the mango represents:

- (1) mesocarp.
- (2) epicarp.
- (3) endosperm.
- (4) endocarp.

148. Which of the following is **not** a part of a leaf?

- (1) Pedicel
- (2) Leaf base
- (3) Petiole
- (4) Lamina

149. Which of the following is the **correct** floral formula for mustard plant?

- (1) $\oplus \overline{\text{♀}} K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$
- (2) $\text{Br} \oplus \overline{\text{♀}} \overbrace{P_{(3+3)}}^{\text{♂}} A_{3+3} G_{(3)}$
- (3) $\oplus \overline{\text{♀}} K_{(5)} \overbrace{C_{(5)}}^{\text{♂}} A_5 \underline{G}_{(2)}$
- (4) $\% \overline{\text{♀}} K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Artificial system of classification brought together the closely related species.

Reason R: Artificial systems gave equal weightage to vegetative and sexual characteristics.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is not the correct explanation of A.

SECTION-(IV) ZOOLOGY**SECTION - A**

- 151.** Which of the following is the basic structural and functional unit of skeletal muscle?
- (1) Myofibril
 - (2) Sarcomere
 - (3) Muscle fiber
 - (4) Myosin
- 152.** The first seven rib pairs, called true ribs, attach dorsally to X vertebrae and ventrally to the sternum via Y .
- Choose the option which fills the blank **correctly**.
- (1) X- thoracic, Y- hyaline cartilage
 - (2) X- thoracic, Y- elastic cartilage
 - (3) X- cervical, Y- elastic cartilage
 - (4) X- cervical, Y- hyaline cartilage
- 153.** All of the following statements are correct w.r.t skeletal system **except**:
- (1) Bone and cartilage are specialised connective tissues.
 - (2) Bone has a very hard matrix due to calcium salts.
 - (3) Cartilage has pliable matrix due to chondroitin salts.
 - (4) The facial region is made up of 22 skeletal bone.
- 154.** Which of the following type of muscle is found in the heart?
- (1) Smooth muscle (2) Skeletal muscle
 - (3) Cardiac muscle (4) All of these
- 155.** Which of the following bone includes the acromion process?
- (1) Humerus (2) Ulna
 - (3) Scapula (4) Sternum
- 156.** Given below are two statements:
- Statement I:** Each vertebra has a central hollow portion (neural canal) through which the spinal cord passes.
- Statement II:** First vertebra is the axis and it articulates with the occipital condyles
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 157.** The light bands of skeletal muscles are known as:
- (1) isotropic bands. (2) anisotropic bands.
 - (3) intercalated disc. (4) cross bridges.
- 158.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** Joints are points of contact between bones only.
- Reason R:** Joint acts as a fulcrum during movement.
- In the light of the above statements, choose the correct answer from the options given below.
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
- 159.** Which of the following enzyme breaks down acetylcholine at the neuromuscular junction leading to muscle relaxation?
- (1) Acetylcholinesterase
 - (2) ATPase
 - (3) Lipase
 - (4) Cholinase
- 160.** Which of the following disease is a result of accumulation of uric acid crystals?
- (1) Osteoporosis (2) Arthritis
 - (3) Tetany (4) Gout
- 161.** Given below are two statements:
- Statement I:** Muscular dystrophy is a genetic disorder.
- Statement II:** Scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the fifth ribs.
- In the light of the above statements, choose the most appropriate answer from the options given below.
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 162.** The joint between the carpal and the metacarpal of thumb is an example of:
- (1) hinge joint. (2) ball and socket joint.
 - (3) saddle joint. (4) pivot joint.

163. Given below are two statements:

Statement I: The central part of thick filament, not overlapped by thin filaments is called the 'H' zone.

Statement II: The thick filaments in the A band are held together by the M line.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

164. What is the function of synovial fluid?

- (1) Lubrication of joints
- (2) Shock absorption
- (3) Nutrient supply to cartilage
- (4) All of these

165. Complete the analogy and choose the **correct** option w.r.t each human limb.

Metatarsals: 5 :: tarsals : _____

- (1) 4 (2) 5
- (3) 6 (4) 7

166. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A subunit of troponin blocks myosin-binding sites on actin in a relaxed skeletal muscles.

Reason R: In the absence of calcitonin, tropomyosin covers the binding sites, preventing cross-bridge formation and contraction.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

167. The U-shaped bone at the base of the buccal cavity is:

- (1) hyoid. (2) ethmoid.
- (3) sphenoid. (4) pterygoid.

168. While of the following is the role of myoglobin in muscle cells?

- (1) Provide energy
- (2) Store oxygen
- (3) Facilitate contraction
- (4) Produce ATP

169. Read the following statements (A-E).

- A. Ribs have two articulation surfaces at the dorsal end, making them bicephalic.
- B. Pubic symphysis is comprising of fibrous cartilage.
- C. Last 2 pairs of ribs are called floating ribs.
- D. Myasthenia gravis is an autoimmune disorder.
- E. The number of mitochondria are few in red muscle fibres.

How many of the above statements is/are **correct**?

- (1) One (2) Two
- (3) Three (4) Four

170. Which of the following cells do **not** exhibit amoeboid movement?

- (1) Macrophages (2) Neutrophils
- (3) Basophils (4) RBCs

171. Which of the following joints is found in between humerus and pectoral girdle?

- (1) Hinge joint (2) Ball-and-socket joint
- (3) Pivot joint (4) Saddle joint

172. Which of the following muscle types is striated?

- (1) Smooth muscle (2) Cardiac muscle
- (3) Skeletal muscle (4) Both (2) and (3)

173. What happens during muscle relaxation?

- (1) Calcium ions are released.
- (2) Calcium ions are pumped back into the sarcoplasmic reticulum.
- (3) Myosin heads bind to actin.
- (4) ATP is consumed.

174. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The contraction of muscles is due to the sliding filament mechanism.

Reason R: During muscle contraction, actin filaments slide over myosin filaments.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Ciliary movement	(I)	ATP binding sites
(B)	Myosin	(II)	Female reproductive tract
(C)	Myoglobin	(III)	Inflammation of joints
(D)	Arthritis	(IV)	Present in red muscle fibers

Choose the **correct** answer from the options given below.

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-III, C-II, D-I

176. Which of the following muscles are responsible for the movement of the skeleton?

- (1) Cardiac muscles
- (2) Smooth muscles
- (3) Skeletal muscles
- (4) All of these

177. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The vertebral column is made up of vertebrae that are separated by intervertebral discs.

Reason R: Almost all mammals, including human beings have five cervical vertebrae.

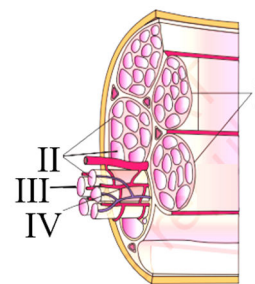
In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

178. Which of the following is a characteristic of smooth muscles?

- (1) They are involuntary.
- (2) They are found in the inner wall of alimentary canal.
- (3) They are striated.
- (4) Both (1) and (2)

179. Refer to the given below diagram and choose the **correct** option.



- (1) I-Fascicle
- (2) II-Sarcolemma
- (3) III-Muscle fibre
- (4) IV-Lymph capillary

180. Which of the following disorders is caused due to low Ca^{++} in body fluid?

- (1) Tetany
- (2) Osteoporosis
- (3) Rheumatoid arthritis
- (4) Myasthenia gravis

181. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pseudopodia	(I)	Fibrous joint
(B)	Actin	(II)	Amoeboid movement
(C)	Cranial bones	(III)	Thin filament
(D)	Femur	(IV)	Longest bone

Choose the **correct** answer from the options given below.

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-I, C-II, D-IV

182. Which of the following options is **correct** w.r.t function of vertebral column?

- (1) It protects spinal cord.
- (2) It supports the head.
- (3) It serves as a point of attachment for the ribs.
- (4) All of these

183. Choose the **incorrect** match.

- (1) Coxal bone – Pelvic girdle
- (2) Ulna - Appendicular skeleton
- (3) Sternum - Axial skeleton
- (4) Humerus - Pelvic girdle

184. Select the **correct** option.

- (1) HMM = Tail + Short arm
- (2) LMM = Tail + Head
- (3) HMM = Head + Short arm
- (4) LMM = Head

- 185.** Read the following statements (A-D).
- A. Patella is a cup shaped bone.
 - B. Myoglobin is a red coloured oxygen storing pigment.
 - C. A motor neuron alongwith the muscle fibres connected to it constitute a motor unit.
 - D. Amount of sarcoplasmic reticulum is high in white muscle fibres.

Which of the above statements are **correct**?

- (1) A and D only
- (2) A, B and C only
- (3) C and D only
- (4) A, B, C and D

SECTION - B

- 186.** Which of the following is **incorrect** w.r.t skeletal muscles?

- (1) They have striped appearance.
- (2) They are voluntary muscles.
- (3) They are primarily involved in locomotory actions.
- (4) They are found in internal hollow organs.

- 187.** Stimulation of a skeletal muscle fibre by a motor neuron occurs at:

- (1) the neuromuscular junction.
- (2) the transverse tubules.
- (3) the myofibril.
- (4) the sarcoplasmic reticulum.

- 188.** Given below are two statements:

Statement I: Muscle contraction is initiated by a signal sent by the peripheral nervous system.

Statement II: The junction between a motor neuron and the sarcolemma of the muscle fibre is called the motor-end plate.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

- 189.** Sarcoplasmic reticulum of skeletal muscle fibres is a storehouse of:

- (1) Ca^{2+} .
- (2) Na^{+} .
- (3) K^{+} .
- (4) Fe^{2+} .

- 190.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Dust particle inhaled along with the atmospheric air can be removed from trachea.

Reason R: Trachea is lined with ciliated epithelium.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

- 191.** Middle ear contains three tiny bones:

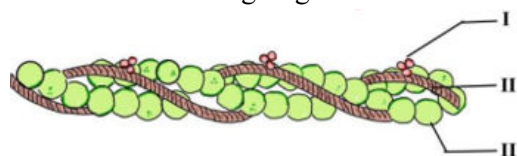
- (1) stapes.
- (2) malleus.
- (3) incus.
- (4) all of these

- 192.** Acetabulum is a cavity to which _____ bone articulates.

Choose the **correct** option to fill the blank.

- (1) thigh bone
- (2) fibula
- (3) sacrum
- (4) tibia

- 193.** Refer to the following diagram



Which of the above represents tropomyosin?

- (1) I
- (2) II
- (3) III
- (4) Both (1) and (2)

- 194.** Select the **correct** option.

- (1) Upper arm: Striated and involuntary
- (2) Fallopian tubes: Non-striated and involuntary
- (3) Heart: Non-striated and involuntary
- (4) Alimentary canal - Striated and involuntary

- 195.** All of the following are correct statements, **except**:

- (1) Many monomeric proteins called meromyosins constitute one thick filament.
- (2) The globular head of myosin has an active ATPase enzyme.
- (3) White muscles fibres depend on anaerobic process for energy.
- (4) Vertebral column is formed by 16 serially arranged units called vertebrae.

196. Select the **incorrect** statement.

- (1) I-band gets reduced during contraction.
- (2) A-band retains length during contraction.
- (3) Both actin and myosin protein are arranged parallel to each other as rod-like structures.
- (4) Muscles with very high quantity of haemoglobin appear pale or whitish.

197. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Cranial bones	(I)	2
(B)	Forelimb	(II)	8
(C)	Ribs	(III)	30
(D)	Fibula	(IV)	24

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III D-IV
- (2) A-III, B-IV, C-I D-II
- (3) A-II, B-III, C-I D-IV
- (4) A-II, B-III, C-IV D-I

198. Read the following statements (A-E).

- A. Each half of pectoral girdle consists of a clavicle and a scapula.
- B. The 8th, 9th and 10th pairs of ribs do not articulate directly with the sternum.

- C. Glenoid cavity articulates with the head of humerus.
- D. Human skull is monocondylic.
- E. The two halves of pelvic girdle meet dorsally to form pubic symphysis.

Which of the above statements are **correct**?

- (1) A, B and C only
- (2) A, C and D only
- (3) A, B, C and D only
- (4) A, B, C, D and E

199. Select the **incorrect** statement.

- (1) Muscle fibre is lined by the plasma membrane called sarcoplasmic reticulum.
- (2) About 40-50 % of the body weight is contributed by muscles.
- (3) Skeletal muscle fibre is a syncytium as the sarcoplasm contains many nuclei.
- (4) Clavicle is commonly called the collar bone.

200. Each coxal bone is formed by the fusion of:

- (1) ilium.
- (2) ischium.
- (3) pubis.
- (4) all of these

